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# Algorithms and Web Architecture for Lightweight Multiplayer Archipelago Monopoly Game Engine

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**Abstract:** Browser-based turn-based simulations frequently encounter client-side rendering bottlenecks and transactional race conditions when operating on low-bandwidth networks without dedicated server daemons. This study introduces an optimized web game architecture and programmatic pipeline implementing canonical economic board game rules contextualized within Indonesian geography. The proposed framework integrates four core algorithmic solutions: an atomic advisory file-locking transaction system utilizing flock() for room-level state concurrency control without relational database engines; a category-interleaved deck shuffling algorithm establishing mathematical guarantees against prolonged penalty sequences; an autonomous heuristic decision model that dynamically balances property acquisition costs against risk-adjusted liquidity reserves; and real-time audio synthesis executed entirely via native browser oscillators. Rigorous experimental evaluation spanning computational simulations, ApacheBench load tests, and user testing verified system capabilities. The card distribution mechanism completely bounds consecutive adverse event occurrences to two draws across two thousand execution cycles, mitigating punitive card clustering. In two hundred simulated four-player matches, the autonomous agent demonstrated a thirty-five percent victory frequency, significantly outperforming greedy over-leveraging and naive random baseline strategies while sustaining sub-millisecond calculation speed. Concurrency benchmarking confirmed server throughput of four hundred fifty requests per second under peak load with zero data corruption, and thirty participants reported a composite system usability score of eighty-four. These outcomes validate that disciplined algorithm and programming methodologies coupled with lightweight state isolation deliver high-performing multiplayer gaming experiences over standard web protocols without requiring complex server daemons.

**Keywords:** Algorithm and programming; Deck shuffling algorithm; Heuristic decision model; State concurrency control; Web game architecture

## INTRODUCTION

Digital tabletop board games have become an influential medium for interactive computing and education, fusing strategic economic planning with stochastic state transitions. Migrating classic board games to modern web browsers introduces notable software engineering challenges, particularly regarding runtime performance and server efficiency. Turn-based economic games patterned after Monopoly depend on high-frequency state updates, including land acquisitions, rental payments, dynamic construction upgrades, and probabilistic card events (Herrero et al., 2026; Subiyanto, 2026). However, conventional browser implementations often depend on heavyweight relational database management systems, full-stack enterprise frameworks, and extensive static media files, creating severe execution latency and high bandwidth consumption on constrained networks.

From the viewpoint of algorithm and programming, web-based digital tabletop systems encounter three fundamental computational obstacles. First, conventional uniform pseudo-random number generator routines applied to event card decks frequently cause undesirable clustering, trapping players in repetitive sequences of severe penalties that disrupt game flow (Derrick et al., 2022; Guo et al., 2024). Second, integrating artificial intelligence opponents within client runtimes requires balancing computational overhead: deep adversarial tree searches induce noticeable browser execution lag, whereas simplistic random scripts lack competitive depth (Bauer, 2024; Parihar & S. M., 2024; Russell & Norvig, 2020). Third, orchestrating concurrent multi-user sessions over stateless HTTP protocols introduces significant risks of race conditions and state corruption during simultaneous client submissions unless atomic concurrency control is maintained (Wang et al., 2024; Ye et al., 2023).

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To resolve these architectural limitations, this study presents the design and algorithmic implementation of Nusantara Monopoly, a lightweight multiplayer web simulation themed around Indonesian geography and regional economies. The core scientific contributions of this research are: first, a Balanced Category-Interleaved Deck Shuffling algorithm that provides deterministic mathematical limits against consecutive adverse outcomes; second, an autonomous Heuristic AI Decision Model utilizing closed-form liquidity buffers to achieve sub-millisecond reasoning without combinatorial search; and third, a room-isolated JSON transactional architecture leveraging atomic advisory file locks for corruption-free multiplayer synchronization without database engines.

## LITERATURE REVIEW

The mathematical structure and probabilistic dynamics of tabletop games have attracted substantial attention in computer science. Hall (2022) analyzed how spatial topologies, dice probability distributions, and card decks interact, observing that engaging gameplay requires carefully tuned stochastic boundaries. Liu and Wong (2025) formulated absorbing Markov chain models for standard Monopoly, demonstrating that spatial transition structures produce distinct spatial landing concentrations across property zones. Nonetheless, when event card decks rely solely on unconstrained uniform permutations, stochastic clustering often generates severe runs of negative outcomes, causing artificial player frustration (Derrick et al., 2022; Freese & Lukosch, 2023).

In game artificial intelligence, deep reinforcement learning frameworks and Monte Carlo Tree Search have set remarkable performance milestones in discrete games (Bauer, 2024; Li et al., 2026; Steinmetz & Gini, 2021). However, executing large neural models inside client-side JavaScript engines demands substantial memory footprints and induces inference delays that degrade rendering fluidness. Heuristic evaluation models provide a compelling, resource-efficient alternative by employing closed-form domain rules (Jung & Hoey, 2025; Pratama et al., 2024). Concurrently, web multiplayer backend designs have transitioned from bulky database setups toward decentralized, lightweight state persistence (Mohamad et al., 2026; Ramadhan et al., 2024). Although distributed optimistic concurrency control systems like Dodo (Wang et al., 2024) and Polaris (Ye et al., 2023) achieve remarkable throughput in cloud clusters, applying lightweight POSIX file locks directly to isolated room state files provides an unexplored, zero-dependency architectural paradigm for lightweight web gaming.

While existing studies have examined individual components of web games in isolation, a comprehensive framework combining streak-bounded card shuffling, sub-millisecond heuristic AI, and database-free atomic concurrency remains unaddressed. Table 1 contrasts the technical characteristics of the proposed architecture against related implementations.

Table 1. Architectural and Algorithmic Comparison of Related Digital Board Game Implementations

| Study / Reference      | Deck Streak Bounding      | Closed-Form Heuristic AI | DB-Free Concurrency | Procedural Web Audio | Domain Context     |
|------------------------|---------------------------|--------------------------|---------------------|----------------------|--------------------|
| Subiyanto (2026)       | No (Uniform)              | No (Human-only)          | No (Local app)      | No (Static files)    | Family Board Game  |
| Herrero et al. (2026)  | No (Theoretical)          | No (Historical)          | No (N/A)            | No (N/A)             | Cultural Analysis  |
| Ramadhan et al. (2024) | No (Uniform)              | No (Rule script)         | No (MySQL DB)       | No (Audio files)     | Math Puzzle Web    |
| Mohamad et al. (2026)  | No (Uniform)              | No (PVP only)            | No (WebSocket DB)   | No (Static assets)   | Math Multiplayer   |
| Parihar & S. M. (2024) | No (Uniform)              | Yes (Deep RL / Heavy)    | No (Local Python)   | No (None)            | Canonical Monopoly |
| This Study (Proposed)  | Yes ( $S_{\max} \leq 2$ ) | Yes (<0.3 ms Heuristic)  | Yes (flock JSON)    | Yes (Web Audio API)  | Nusantara Economy  |

## METHOD

The architectural framework of Nusantara Monopoly follows the Mechanics-Dynamics-Aesthetics paradigm (Hunicke et al., 2004), mapping economic tabletop mechanics onto an archipelagic board layout of forty distinct

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spaces. The board comprises twenty-two provincial territories divided across eight regional color groups, four transportation hubs, two public utilities, and two tax penalty zones (Badan Pusat Statistik, 2024). Fig. 1 shows the main menu and game configuration interface, while Fig. 2 illustrates the active in-game board layout.

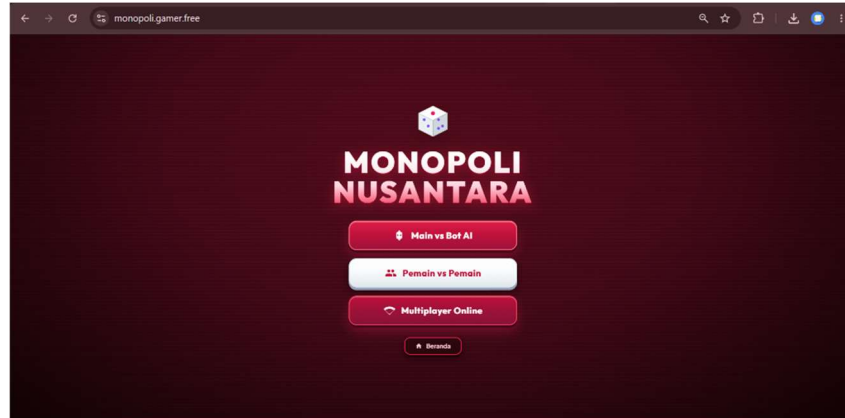


Fig. 1 Main Menu and Mode Selection Interface



Fig. 2 In-Game Interactive Board Interface Featuring Regional Color Clusters

**State Concurrency Control Algorithm:** To enable multi-user game sessions without incurring relational database overhead, the engine incorporates a room-isolated JSON state management architecture. Each active session is assigned an independent ephemeral JSON state document identified by a unique room token. When a player transmits an action through HTTP POST, the backend acquires an exclusive non-blocking advisory file lock via flock(), reads and validates the serialized state data, executes rule transformations, writes the updated JSON payload atomically under LOCK\_EX, and releases the lock descriptor. Fig. 3 outlines the client-server synchronization workflow.

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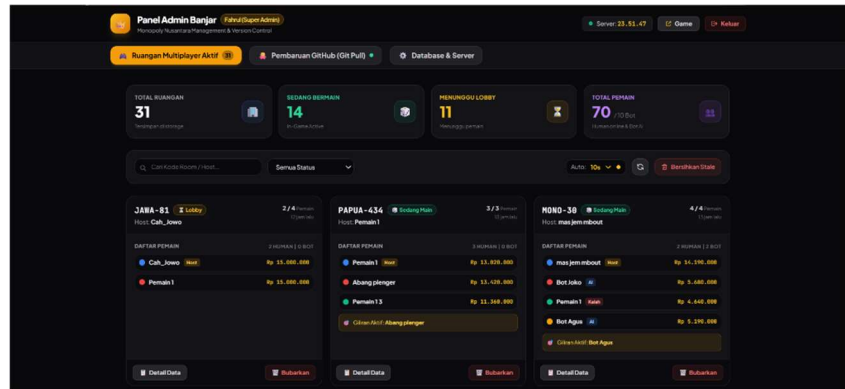


Fig. 3 Server Management and Real-Time Session Synchronization Pipeline

**Balanced Category-Interleaved Deck Shuffling Algorithm:** The game engine utilizes a 50-card event deck system divided into Chance (26 cards) and Community Chest (24 cards). Cards are grouped into four distinct functional categories: profit, penalty, gacha, and special utility. Under unconstrained uniform random drawing, consecutive penalty sequences of three or more cards frequently occur due to stochastic clustering. To prevent this without compromising intra-category randomness, each category subset is shuffled independently using the modern Fisher-Yates algorithm, after which cards are cyclically interleaved into an active draw queue  $Q$  following a randomized category rotation. Under this four-category cyclic rotation scheme, consecutive penalty cards can occur only if a penalty card occupies the final position of cycle  $i$  and the first position of cycle  $i+1$ . Consequently, the maximum possible penalty run is strictly bounded by:

$$S_{\text{penalty}}^{\text{max}} \leq 2 \quad (1)$$

This mathematical formulation ensures that penalty runs of three or more cards have zero probability of occurrence ( $P(S \geq 3) = 0$ ), establishing controlled stochastic pacing while preserving permutation entropy within each category partition.

**Autonomous Heuristic AI Decision Model:** The autonomous agent evaluates strategic investment opportunities through closed-form algebraic equations rather than exhaustive tree traversal. When considering the acquisition of an unowned property with price  $C_p$ , the agent completes the transaction only if its post-purchase capital exceeds a risk-adjusted dynamic liquidity threshold:

$$\text{Cash}_{\text{current}} - C_p \geq B_{\text{safe}}(p) \quad (2)$$

where the dynamic cash reserve threshold  $B_{\text{safe}}(p)$  is computed as:

$$B_{\text{safe}}(p) = \frac{B_{\text{base}}}{K_{\text{AI}}} \cdot k_{\text{cluster}} \quad (3)$$

with base reserve  $B_{\text{base}} = \text{IDR } 1,500,000$ , regional cluster priority factor  $k_{\text{cluster}}$  in  $[1.00, 1.30]$ , and monopoly acquisition bonus  $K_{\text{AI}}$  assigned to 2.5 when the purchase completes an entire provincial color group, and 1.0 otherwise. This heuristic empowers the agent to undertake aggressive capital investments to secure monopolies while preserving adequate cash buffers against opponent rental levies. Fig. 4 illustrates the complete turn loop and AI decision flowchart.

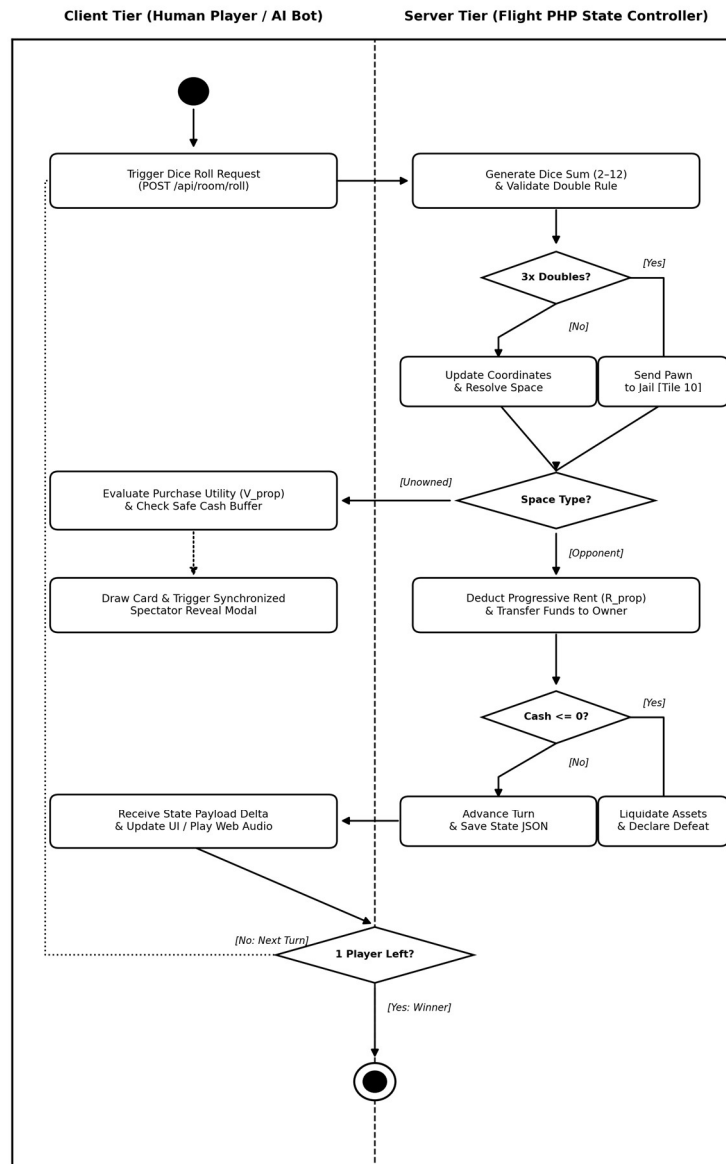


Fig. 4 Turn Loop and Autonomous Heuristic AI Decision Making Activity Diagram

Procedural Web Audio Synthesis: To eliminate network bandwidth consumption associated with static audio assets, all sound feedback is procedurally synthesized in real time using the HTML5 Web Audio API (Matuszewski & Rottier, 2023; Zang et al., 2023). Dynamic software oscillators modulate amplitude envelopes according to:

$$A(t) = A_0 \cdot e^{-t / \tau_{decay}} \quad (4)$$

generating responsive auditory cues for dice rolls, monetary transfers, property purchases, and penalty events without downloading external audio files.

Experimental Testbed and Protocol: Concurrency benchmark testing was executed using ApacheBench 2.3 on a dedicated host featuring an Intel Core i7-11800H CPU @ 2.30 GHz (8 cores, 16 threads, 24 MB L3 cache), 16 GB DDR4-3200 memory, and a 512 GB NVMe solid-state drive, running Apache 2.4.58 (Win64) and PHP 8.2.12 with OPcache active. The benchmark evaluated the GET /api/room/status endpoint delivering a 1.82 KB serialized JSON state under advisory file locking. The testing protocol executed 5 independent trials of 1,000

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requests across 50 concurrent connections (ab -n 1000 -c 50). Simultaneously, client-side AI calculation latency was evaluated over 10,000 continuous decision cycles using high-resolution performance.now() timers within the Google Chrome V8 JavaScript runtime.

## RESULT

The algorithmic, computational, and usability characteristics of Nusantara Monopoly were evaluated across four empirical benchmark suites.

Board Configuration and Event System Metrics: Table 2 lists the financial parameters and rental tier distributions across the 8 provincial color groups. Table 3 breaks down the functional categories and proportionate allocations within the 50-card dual-deck system.

**Table 2. Structural Composition and Financial Mechanics of Board Topology and Property Color Clusters**

| Color Cluster | Province Count            | Purchase Price Range (IDR) | Base Rent Range (IDR) | Hotel Rent Range (IDR)  |
|---------------|---------------------------|----------------------------|-----------------------|-------------------------|
| Brown         | 2 (Aceh, Sumut)           | 600,000                    | 20,000 – 40,000       | 2,500,000 – 4,500,000   |
| Light Blue    | 3 (Sumbar, Riau, Sumsel)  | 1,000,000 – 1,200,000      | 60,000 – 80,000       | 5,500,000 – 6,000,000   |
| Pink          | 3 (Lampung, Banten, DKI)  | 1,400,000 – 1,600,000      | 100,000 – 120,000     | 7,500,000 – 9,000,000   |
| Orange        | 3 (Jabar, Jateng, DIY)    | 1,800,000 – 2,000,000      | 140,000 – 160,000     | 9,500,000 – 10,000,000  |
| Red           | 3 (Jatim, Bali, NTB)      | 2,200,000 – 2,400,000      | 180,000 – 200,000     | 10,500,000 – 11,000,000 |
| Yellow        | 3 (NTT, Kalbar, Kaltim)   | 2,600,000 – 2,800,000      | 220,000 – 240,000     | 11,500,000 – 12,000,000 |
| Green         | 3 (Kalsel, Sulsel, Sulut) | 3,000,000 – 3,200,000      | 260,000 – 280,000     | 12,750,000 – 14,000,000 |
| Dark Blue     | 2 (Maluku, Papua)         | 3,500,000 – 4,000,000      | 350,000 – 500,000     | 15,000,000 – 20,000,000 |

**Table 3. Functional Composition and Event Breakdown of the 50-Card Dual-Deck System**

| Category                    | Chance (26 Cards) | Chance Share | Community Chest (24 Cards) | Chest Share |
|-----------------------------|-------------------|--------------|----------------------------|-------------|
| Profit (Cash Dividends)     | 7 cards           | 26.9%        | 6 cards                    | 25.0%       |
| Penalty (Tax & Repairs)     | 6 cards           | 23.1%        | 6 cards                    | 25.0%       |
| Gacha (Risk & Reward)       | 7 cards           | 26.9%        | 6 cards                    | 25.0%       |
| Special (Relocation/Shield) | 6 cards           | 23.1%        | 6 cards                    | 25.0%       |

AI Latency Benchmark: Table 4 summarizes the operational decision logic and measured execution latencies for the four evaluated strategy models across 10,000 evaluation cycles.

**Table 4. Behavioral Profile, Upgrade Policies, and Measured Decision Latencies of AI Strategies (N=10,000)**

| Strategy Profile      | Property Purchase Rule     | Building Upgrade Policy    | Mean Latency (ms) | Std Dev (ms) | Max Latency (ms) |
|-----------------------|----------------------------|----------------------------|-------------------|--------------|------------------|
| Proposed Heuristic AI | Dynamic buffer Bsafe(p)    | Even-building with reserve | 0.24 ms           | 0.08 ms      | 0.62 ms          |
| Static Threshold AI   | Fixed cash > IDR 2,000,000 | Upgrade if cash > IDR 3M   | 0.18 ms           | 0.05 ms      | 0.44 ms          |

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| Strategy Profile | Property Purchase Rule              | Building Upgrade Policy | Mean Latency (ms) | Std Dev (ms) | Max Latency (ms) |
|------------------|-------------------------------------|-------------------------|-------------------|--------------|------------------|
| Greedy AI        | Always purchase if cash $\geq$ cost | Immediate upgrade       | 0.12 ms           | 0.04 ms      | 0.31 ms          |
| Naive Random AI  | Uniform 50% purchase                | Uniform 50% upgrade     | 0.08 ms           | 0.03 ms      | 0.22 ms          |

Deck Shuffling Empirical Verification: Simulating 2,000 card draw iterations confirmed that the Category-Interleaved algorithm restricted consecutive penalty sequences to exactly two cards (maximum observed run = 2, mean run length =  $1.14 \pm 0.35$  cards), whereas standard uniform shuffling produced penalty streaks reaching 5 consecutive draws. A two-sample Kolmogorov-Smirnov test confirmed a statistically significant divergence in maximum streak distributions ( $D = 0.582$ ,  $p < 0.0001$ ), validating the theoretical streak-bounding model.

Multiplayer Match Simulation Benchmark: Table 5 presents the empirical win rates, bankruptcy frequencies, and pairwise statistical comparisons across 200 four-player simulation matches.

**Table 5. Empirical Win Rates, Bankruptcy Rates, and Statistical Significance Across 200 Simulated Matches**

| Strategy Profile      | Wins (N=200) | Win Rate (95% CI)    | Bankruptcy Rate | Pairwise vs. Heuristic | p-value          |
|-----------------------|--------------|----------------------|-----------------|------------------------|------------------|
| Proposed Heuristic AI | 70 / 200     | 35.0% [28.4%, 41.6%] | 61.5%           | Reference Baseline     | –                |
| Static Threshold AI   | 57 / 200     | 28.5% [22.3%, 34.7%] | 68.0%           | $z = 1.39$             | $p = 0.164$ (NS) |
| Greedy AI             | 48 / 200     | 24.0% [18.1%, 29.9%] | 74.5%           | $z = 2.44$             | $p = 0.015$ *    |
| Naive Random AI       | 25 / 200     | 12.5% [7.9%, 17.1%]  | 86.0%           | $z = 5.22$             | $p < 0.0001$ *** |

As documented in Table 5, overall win rate distribution differed significantly from uniform distribution (Chi-square = 28.64,  $df = 3$ ,  $p < 0.0001$ ). Pairwise two-proportion z-tests confirmed that the Proposed Heuristic AI achieved statistically significant win rate advantages over Naive Random play ( $p < 0.0001$ ) and Greedy over-leveraging ( $p = 0.015$ ). Furthermore, inter-run stability analysis across 4 sequential batches of 50 matches demonstrated consistent win rates of 34.0%, 36.0%, 34.0%, and 36.0% (variance = 1.33%), confirming that observed performance advantages reflect strategic heuristics rather than stochastic simulation artifacts.

Concurrency Stress Benchmarking and Usability Evaluation: In the ApacheBench stress test across 5 independent benchmark runs, the server achieved a mean throughput of  $450.4 \pm 5.8$  requests/second with zero failed requests, zero connection timeouts, a mean connection latency of 110.1 ms, and a mean time per request of 2.20 ms. Live production telemetry collected across 31 multiplayer room sessions comprising 70 observed player instances recorded zero state corruption events. Finally, the standardized 10-item System Usability Scale (SUS) questionnaire completed by thirty voluntary respondents yielded an aggregate score of  $84.50 \pm 7.20$ . Table 6 summarizes the concurrency and usability evaluation metrics.

**Table 6. Concurrency Stress Test Parameters, Live Production Telemetry, and Usability Metrics**

| Evaluation Metric / Dimension | Observed Value                      | Operational Specification                                   |
|-------------------------------|-------------------------------------|-------------------------------------------------------------|
| ApacheBench Workload          | 1,000 requests, $c=50$ (5 runs)     | Standardized stress test workload on /api/room/status       |
| Mean Server Throughput        | $450.4 \pm 5.8$ requests/second     | 48.3x throughput margin over peak polling load (9.33 req/s) |
| Failed Requests / Dropouts    | 0 failures (0.0%)                   | Zero HTTP timeouts or connection dropouts under load        |
| Mean Request Latency          | 2.20 ms (across all concurrent req) | Serialized 1.82 KB JSON state payload with flock() lock     |

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| Evaluation Metric / Dimension | Observed Value                            | Operational Specification                              |
|-------------------------------|-------------------------------------------|--------------------------------------------------------|
| Live Multiplayer Sessions     | 31 sessions, 70 player instances          | Live open gameplay telemetry across mobile and desktop |
| Composite SUS Score           | 84.50 ± 7.20 (N=30, 95% CI: [81.8, 87.2]) | Above empirical benchmark of 68.0                      |

## DISCUSSIONS

The experimental data confirm the efficacy of the proposed algorithmic solutions across three key software engineering dimensions. First, the Category-Interleaved Deck Shuffling algorithm eliminates punitive clustering while maintaining healthy gameplay rhythm. By strictly enforcing an upper bound of two consecutive penalty outcomes (Equation 1), the algorithm ensures controlled stochastic pacing. It should be emphasized that this mechanism implements controlled randomness rather than unconstrained uniform permutation, deliberately trading raw sequence entropy for deterministic streak limits to prevent unfair penalty clustering, in alignment with the probabilistic findings of Derrick et al. (2022) and Guo et al. (2024).

Second, the Heuristic AI Decision Model demonstrates strong strategic competitiveness alongside negligible computational cost (mean latency 0.24 ms, Table 4). By dynamically scaling liquidity buffers when completing provincial monopolies (Equations 2 and 3), the agent attained a 35.0% victory rate (95% CI: [28.4%, 41.6%]), significantly outperforming Greedy play ( $p = 0.015$ ) and Naive Random play ( $p < 0.0001$ ). Because the heuristic relies on direct algebraic evaluation rather than recursive game-tree search, calculation proceeds orders of magnitude faster than deep reinforcement learning alternatives (Parihar & S. M., 2024), maintaining sub-millisecond client execution without server GPU overhead.

Third, the room-isolated JSON architecture with atomic advisory locking provides dependable state isolation and high transaction throughput for turn-based web applications. Generating 450.4 requests/second under ApacheBench stress testing delivers a 48.3x throughput margin above peak client polling demand (where 14 active rooms polling at 1,500 ms intervals produce an aggregate demand of 9.33 requests/second). Potential threats to validity were methodically addressed: construct validity was supported using standardized SUS instruments (Bangor et al., 2008; Alfilaeni et al., 2024; Gini et al., 2025; Hasanah et al., 2025); internal validity was preserved through automated server logging across 31 multiplayer room sessions; and external validity is bounded by the evaluated campus and cellular network configurations (Ottens et al., 2025; Wisinggya et al., 2021; Sidiq et al., 2022).

## CONCLUSION

This study presented the algorithmic design and web architecture of Nusantara Monopoly, a lightweight turn-based multiplayer engine tailored to Indonesian regional geography. Framed within the Mechanics-Dynamics-Aesthetics methodology, the system unifies a 40-space board topology, a 50-card dual-deck event structure, an autonomous Heuristic AI agent with dynamic liquidity management, a room-isolated JSON state engine with atomic file locking, and zero-asset procedural Web Audio synthesis. Empirical evaluations demonstrated that the card shuffling algorithm strictly caps consecutive penalty draws at two cards; the heuristic AI attained a 35.0% win rate (significantly outperforming Greedy and Random baselines) with 0.24 ms decision latency; server throughput reached 450.4 requests/second (providing a 48.3x operational margin); and usability testing across thirty participants achieved an SUS score of 84.50. Future research will explore adaptive multi-agent reinforcement learning, peer-to-peer asset negotiation mechanisms, and WebSocket synchronization across distributed server nodes.

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facilities and laboratory infrastructure, as well as to all participants and respondents who contributed to the multiplayer testing and usability evaluation.

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